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QoE-aware Q-learning resource allocation for NOMA wireless multimedia communications

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Abstract: Hybrid control traffic and multimedia flow over emerging non-orthogonal multiple access (NOMA) could provide both very low latency and very high bandwidth. In this study, a Q-learning-based resource allocation scheme is proposed to improve the quality of experience (QoE) for NOMA user equipment (UE) in downlink wireless multimedia communications. In the proposed framework, the utility is modelled as the QoE with regard to communication resource cost, where UE acts as the agent in the reinforcement Q-learning. UE observes the wireless channel states and takes resource allocation actions based on the immediate reward of QoE gain and communication cost. In addition, benefiting from the NOMA communications, the authors propose to solve the multiple agent reinforcement learning problems with the simplified sequential single agent reinforcement learning (SARL) approach. The numerical simulation results demonstrate the efficiency of the proposed Q-QoE resource allocation framework and prove that the UE would obtain desirable QoE performance with the SARL scheme.

Nomenclature

N	number of UE in the NOMA system
$\|h_i\|^2$	channel gain of UE$_i$
P_i	allocated power of UE$_i$
σ^2	variance of the channel noise
I_i	SINR experienced by UE$_i$
B	bandwidth of the NOMA system
R_i	achievable data rate of UE$_i$
U_i	QoE gain of UE$_i$
c	UE's unit cost of power
s_t, a_t	agent's state and action at time step t
π	power resource allocation policy
S, A	state space and action space
r_t	reward for an agent by taking a_t at s_t
α	learning rate of the Q-learning
β	discount factor when updating the Q value
ϵ	exploration probability of the learning
$Q(s, a)$	Q value of Q-learning matrix at each state-action

1 Introduction

As the mobile video turns into the dominating traffic in modern internet [1], how to improve quality of experience (QoE) becomes even harder: the huge system capacity, a large number of users, high complexity, and heterogeneous architecture, as well as dramatically increasing demand of resource consumption [2, 3]. In this work, Q-learning is introduced to solve the QoE enhancement issue in wireless multimedia services. More specifically, a multiple agent reinforcement learning (MARL) scenario is studied in our work to model the multiple user equipment (UE) resource allocation activity in the downlink wireless media service. By setting the QoE utility as the agent's reward in the Q-learning process, we could reach the QoE improving goal via obtaining the optimal resource allocation policy of base station (BS).

On the other hand, as the most recent member of the multiple access family in the wireless networks, the non-orthogonal multiple access (NOMA) technology has been exposed to the public and it has been widely studied recently [4]. Owing to its features such as significant spectral efficiency, high user fairness, and allowing the BS serves multiple UE at the same time/frequency/code with different power levels, the NOMA has been proposed for 3rd generation partnership project (3GPP) long-term evolution and considered to be a necessary component in the next generation of wireless network service. Therefore, in our work, we would like to investigate the service scenario where the BS provides media service to multiple UE through the NOMA downlink system, as shown in Fig. 1. The Q-learning is used to help the BS approximate the optimal resource allocation utility, then the BS would provide high service QoE to multiple UE simultaneously.

To the best knowledge of authors, very few papers (if any) introduced the Q-learning method into the multimedia NOMA wireless networks to improve the UE's perceived QoE. The key contributions of this paper are as follows:

(i) To quantitatively analyse UE's perceived QoE in a concave fashion, in this work, we adopted a light-weighted two-level logarithmic mathematical QoE model to present the UE's utility, based on the cost of power resource and wireless channel conditions, such as physical channel gain and interference.
(ii) Multiple UE behaviours could be either competing for the resource allocation for individual QoE gain or cooperating with others for maximising the system energy/throughput efficiency. To properly model and solve the MARL problem in our NOMA downlink system, we translate the MARL to a sequential single-agent reinforcement learning (SARL) scheme to address the multiple UE QoE utility improvement issue.
(iii) A model-free Q-learning approach is developed in our work, through which we would approximate the BS's optimal power resource allocation policy for each UE. The policy would ensure UE to approximate the optimal QoE gain at the same time.
(iv) We present numerical results to illustrate how the system parameters would affect the convergence of the Q-learning algorithm. We also evaluate the system QoE performance enhancement under the proposed SARL scheme in different scales of NOMA downlink users.

The structure of the rest of this paper is as follows. We shortly review the literature of the reinforcement learning methods and the QoE modelling issues in NOMA wireless networks in Section 2. The details of the NOMA downlink model and QoE model for UE are explained in Section 3. How to introduce the reinforcement learning method into our work, and how to use the Q-learning to approximate the optimal resource allocation policy are addressed in

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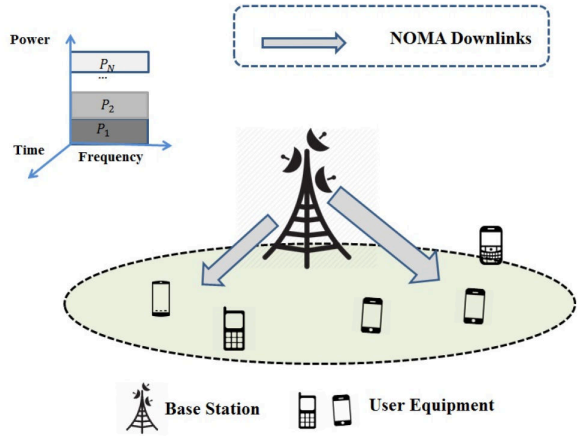


Fig. 1 *Structure of multimedia NOMA downlink wireless service. The BS serves multiple UEs simultaneously with the same time/frequency/code but with different power levels*

details in Section 4. In Section 5, numerical simulations are presented. Finally, we conclude this paper in Section 6.

2 Related work

This paper involves three essential parts: QoE scheduling in wireless multimedia communications, NOMA wireless communications, and machine learning in wireless communications. Therefore, the related works are organised by three categories accordingly.

2.1 QoE scheduling in wireless multimedia communications

To improve the QoE gain of UE in the wireless multimedia services, plenty of research has been conducted in the literature from different aspects, such as resource allocation, transmission scheduling, and system architecture [5–8]. The authors in [5] proposed a user-centric multimedia rebroadcasting scheme in the device-to-device (D2D) communication model by jointly considering energy consumption, modulation rate, and physical channel conditions. Through the cross-layer optimisation, the user's QoE could be dramatically enhanced. In [6], the authors provided an optimal content cache management scheme for hypertext transfer protocol adaptive bit rate streaming over wireless networks. The QoE of the individual content for UE would be maximised under a limited storage budget by solving the convex optimisation problem in two steps. In addition, by taking the space-time coding technology and multimedia content distribution/compression into consideration simultaneously, authors in [7] improved the QoE of UE in the multimedia relay wireless scenario efficiently. In [8], the authors addressed the multiple UE QoE enhancement issue by taking the network coding (NC) technology into consideration. Based on the high transmission efficiency of the NC scheme, their proposed least repetition first transmission algorithm would improve the UE QoE dramatically, even with relatively large numbers of D2D users.

2.2 NOMA wireless communications

In addition, as one of the emerging wireless technologies, NOMA has received lots of attention due to its spectrum efficiency in heterogeneous wireless systems [9–12]. The fundamentals of NOMA include the superposition coding (SC) scheme at the sender and the successive interference cancellation (SIC) scheme at the receiver. For example, in the downlink NOMA transmission, the sender (i.e. the BS) broadcasts the superposition coded signal in the same frequency channel. Then, multiple receivers at different distances (where the SIC process is implemented) would decode the information and remove the interference from all weaker UE. Here, we call the UE with high channel gain as strong UE. The UE with low channel gain is called weak UE. In [9], the authors introduced an overview of the concept, analysed performance evaluation gains, and showed some experimental trials of NOMA to improve the QoE of user and system capacity. Their results also show that NOMA outperformances the orthogonal multiple access in both link-level and system-level. In addition, aiming at the enhancement of system efficiency and QoE of UE, authors in [10] investigated the application of NOMA at the cell edge. By combining with the intra-beam SC and multiple-input multiple-output schemes, the system throughput would be significantly improved in both downlink and uplink scenarios. Authors in [11] proposed a QoE-optimised communication power allocation approach in the downlink video service with NOMA technology. Factors such as the power budget of BS, physical channel constraints, and video packet quality influence are considered simultaneously in their work to improve the QoE of UE in the system. For future radio access in the 2020s-era, the authors in [12] presented the potentials of NOMA technology in the enhancements of spectrum efficiency in lower frequency bands for the 5G downlink system. Multi-user transmission power allocation, scheduling algorithm, receiver design, and multi-antenna NOMA are evaluated with a practical system set up and obtained promising results.

2.3 Machine learning in wireless communications

As one of the most promising artificial intelligence tools, machine learning has one of the focuses on future wireless networks [13–15]. Some conventional optimisation-based or game-theoretical resource assignment schemes are non-deterministic polynomial-time (NP)-hard or too complex to resolve, however, the wireless service could approximate desirable dynamic spectrum access (i.e. frequency bands and times) and resource allocation (i.e. transmission power) through machine learning technology. Jiang *et al*. [14] gave a short review of various types of compelling applications of machine learning in the current and future wireless services, e.g. cognitive radios, femtocell/small-cell, heterogeneous networks, smart grid, energy harvesting, D2D communications, and so on. Where the machine learning is considered as a new paradigm that would allow UE autonomously access all the meritorious resources and thus improve UE QoE. In addition, a detailed comparison of several machine learning algorithms is conducted by researchers in [15]. The authors in that research presented the application of machine learning technologies to improve the congestion control in wireless networks, and eventually showed the interest of machine learning to help designing new protocol in computer networks. Recent research work has been reported in the literature [16], investigated the strength of deep Q-learning in 5G wireless networks. The authors formulated the mapping of multiple service function chains with priorities as a multi-step linear integer programming problem for software-defined networks.

Besides the above-mentioned general utilisation of machine learning, one of its well-known approaches, the reinforcement learning has also gained tremendous popularity and been largely investigated in the modern wireless service architectures, i.e., mobile *ad hoc* networks, D2D communication, and vehicle to vehicle (V2V) communications [17–19]. An overview of reinforcement learning is given in [17], where the events, rules, agent interactions, and coordination of the learning process are investigated in several wireless network schemes. The authors' work is presented in a tutorial manner, to establish a foundation for further research in this field. In [18], one decentralised resource allocation mechanism for the V2V communication system is provided based on the deep reinforcement learning method. The agents in the V2V system would achieve optimal transmission spectrum and power selection decisions based on interacting and learning from the unknown environment. Luo *et al*. [19] studied the Q-learning algorithm under the D2D communication scheme in cellular networks. Their proposed learning method could efficiently solve the radio resource management problem; even prior knowledge is not available.

Referring to the aforementioned work in the literature, we adopt the reinforcement learning idea in our work and developed the epsilon-greedy Q-learning in the multi-user NOMA downlink

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multimedia service scenario. The QoE of UE and the system resource allocation efficiency would be both improved through approximating the optimal action policy generated by the learning process.

3 System models

In this section, we address the utility model of the proposed QoE improvement issue in the NOMA wireless multimedia networks.

Key notations in this paper are summarised in the Nomenclature section.

As shown in Fig. 1, there are N UE (marked as UE_1–UE_N) in the NOMA downlink service and the BS serves them simultaneously. Let $|h_i|^2$ denotes the Rayleigh fading channel gain between the BS and UE_i. In our work, we assume the channel gain of each UE follows non-decreasing order, i.e. $|h_1|^2 \le |h_2|^2 \le \cdots \le |h_N|^2$. Without loss of generality, we assume all downlinks experience the independent and identically distributed (i.i.d.) block Rayleigh fading and additive white Gaussian noise [20, 21].

3.1 Data rate of UE in NOMA downlinks

Benefiting from the SC scheme at the transmitter side, the BS is capable to broadcast the coded data to all UE at the same time. We assume the UE with a stronger channel would perform efficient SIC to cancel signal from weaker UE, i.e. UE_i would channel the signal of UE_j when $i > j$. The signal to interference plus noise ratio (SINR) of UE_i is expressed as

$$I_i = \frac{P_i \times |h_i|^2}{\sum_{k=i+1}^{N} P_k \times |h_k|^2 + \sigma^2}, \forall i \in [1, N-1] \tag{1}$$

where P_i is the transmission power that BS adopted to UE_i and the σ^2 implies the variance of the noise. Equation (1) also indicates that the UE with weak channels would experience interference from strong UE. After the SINR is available, we could obtain the achievable data rate of UE_i according to Shannon law [20, 21]

$$R_i = B \times \log_2(1 + I_i) \tag{2}$$

where B denotes the system bandwidth. Since the UE_N has the strongest channel, it could perform sufficient SIC to obtain its own signal from the channel without interference. Then the achievable data rate of UE_N is calculated as $R_N = B\log_2(1 + P_N \times |h_N|^2)$.

3.2 QoE utility for UE

In our work, the UE's QoE function would be non-decreasing with respect to the transmission rate, and the rate is logarithmic to the power. Hence we use the two-level logarithmic model to describe the QoE gain of UE considering communication power cost, similar to [22, 23]

$$U_i = \log_2(1 + R_i) - c \times P_i \tag{3}$$

s.t.

$$\sum_{i=1}^{N} P_i \le P_{\max}; U_i > 0$$

where the parameter c denotes the unit power cost of UE_i.

This equation describes the utility model of the UE. It is a log function corresponding to the achievable data rate. As we know for most of the image compression and video coding algorithms, typically the larger the data rate, the higher the QoE. However, the relationship between the data rate and the QoE is not linear. When the data rate is increased, the QoE could be significantly improved. This typically represents the main frame structure of a picture or the rough content information of the video. When the data rate is further increased, the QoE will increase but at a lower speed. This typically comes from the fine details of the pictures in the multimedia contents.

From the SINR and data rate calculation equations, i.e. (1) to (2), we conclude that the UE would obtain a high data rate when the BS allocates high transmission power to it. Unfortunately, higher transmission power will cause higher interference to other UE. So we introduce the cost term $c \times P_i$ in the QoE definition formula in (3), which is interpreted as the power cost of UE, to regulate UE's power allocation. The rationale behind this is to make sure the strong UE would not suppress weak UE (e.g. certain UE extremely high power selections would interfere with others badly).

4 Design of reinforcement Q-learning solution

4.1 Reinforcement learning fundamentals

Each UE is an agent in this reinforcement learning approach. The reinforcement learning approach is to find the balance between exploration and exploitation, also known as the trade-off between *uncharted territory* and *current knowledge*. The formal model of reinforcement learning with a single agent is described as a Markov decision process (MDP).

Definition 1: A finite MDP is a tuple $\{S, A, p, r\}$, where S denotes the finite set of environment states, A denotes the finite set of agent actions, p denotes the state transition probability function (i.e. $p: S \times A \times S \to [0, 1]$), and r denotes the reward function when the agent reaches a new state (i.e. $r: S \times A \times S \to r$) [24].

The state $s_t \in S$ implies the entire environment where the agent involved in at each discrete time slot t. The agent observes state s_t and takes an action $a_t \in A$ to next state s_{t+1} based on the transition probability given by p. During the state transition, the agent receives an instant reward according to the reward function. The behaviour of the agent is described by its policy, denoted as π, which specifies the agent how to choose actions at certain given states. The goal of the agent is to find the optimal action policy (i.e. π^*) that would maximise the expected cumulative discounted rewards, from any state s_t

$$G_t = E\left[\sum_{n=0}^{\infty} \beta^n r_{t+n}\right], a_n \in \pi^* \tag{4}$$

where β denotes the discount factor, π^* denotes the optimal action policy for the agent.

The discount factor determines the weight or the importance we put to future rewards in the Q-learning. It shows how much weight we put to the current reward or future rewards. The determination of the Beta is totally out of the scope of this paper. The pure machine learning research could determine what could be a reasonable Beta value for Q-learning. In this paper, we take the Beta as a configuration parameter from the outside.

In our work, reinforcement learning is adopted in the NOMA downlink service to find the optimal power allocation policy in terms of QoE utility. Typically power is treated as a kind of resource in wireless communications, and thus, we use used resource allocation sometimes denoting the power allocation in this paper. Fig. 2 illustrates the structure of the reinforcement learning that we considered in this paper. Different from the conventional MDP, the probability function and reward function are unknown in the NOMA downlink service scenario. So we utilise the model-free reinforcement learning in our scenario to address the optimal power allocation issue. The Q-learning, as one important technique for model-free reinforcement learning, is adopted and discussed in detail in the following context [25].

As depicted in Fig. 2, where an agent, corresponding to an UE in our case, interacts with the environment. At each time t, the agent observes a state s_t from the state space S, takes an action a_t from action space A based on the power allocation policy π, and gains the reward r_{t+1} from the new state s_{t+1}. The state observed by agent for characterising the environment includes three

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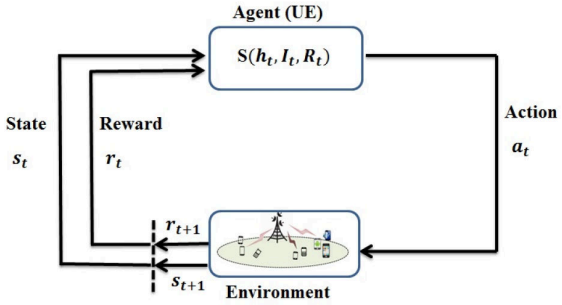


Fig. 2 *Structure of the reinforcement learning model in our work*

components in our model: the instant channel gain of the agent (denoted as h_t), the interference from the NOMA downlink (denoted as I_t), and the data rate of the agent at current state (denoted as R_t). Thus, the state is expressed as $s_t = [h_t, I_t, R_t]$. The action of agent is explained as the power allocation strategy at the next time step (i.e. $a_t = \pi|_{t+1}$). In our work, the reward that the agent gains from taking the action $a_t = \pi|_{t+1}$ is defined as its QoE utility gain under the new power allocation strategy

$$r_t = U_i|_{h_{t+1}, I_{t+1}, R_{t+1}, P_{t+1}}. \quad (5)$$

The objective of the Q-learning is to approximate the optimal power allocation policy for UE that maximises the long-term expected accumulated discounted rewards (i.e. the QoE utility).

4.2 SARL for downlink NOMA multimedia QoE

As shown in Fig. 1, there are N UE in the NOMA downlink system. Each UE acts as an agent in the NOMA system to experience the environment (i.e. channel gain, interference etc.) and take actions individually. Thus, the original goal of designing reinforcement learning is to find the optimal action policy for the multiple agents' system. However, compared with the SARL, the MARL has following challenges [24].

(i) The computational complexity of MARL is bigger than the SARL. The reinforcement learning algorithms (for both MARL and SARL) would estimate values for each possible discrete state or state–action pair, which directly causes the exponential increase of their computational complexity. In addition, the increasing number of agents in MARL and each agent adds its own variables to the joint state–action space would dramatically increase the computational complexity of MARL (as known as severe dimensionality increasing).
(ii) It is difficult to specify the goal of MARL in general stochastic problems. In MARL, the returns (e.g. rewards) of agents are correlated and cannot be maximised independently. More specifically, the stability of agents' learning dynamics and the adaptation to changing behaviour of other agents are two common goals in the MARL research.
(iii) The non-stationarity of the system would dramatically increase in MARL. The reason is that all agents in the system are learning simultaneously, each agent's optimal policy will change as other agents' policies change.
(iv) The need for coordination among agents. Since each agent will be affected by the environment and actions taken by other agents in MARL, the mutual consistency of agents' actions are needed to achieve their intended effect.

To address the stated challenges, we propose a *sequential SARL* method by taking advantage of the feature of NOMA technology, to solve the multi-agent policy optimal problem in our wireless service scenario. To be more specific, we have:

Property 1: To obtain the optimal resource (i.e. power) allocation policy, the MARL problem in the NOMA downlink wireless multimedia service can be simplified as a SARL problem.

Before we show our proof of Property 1, without loss of generality, we have the following assumption.

Assumption 1: The channel condition of UE in the NOMA downlink system would keep consistency during a short service period for one user.

Assumption 1 is based on the rationale that: when BS serves UE simultaneously, the optimal power allocation policy would be adopted until the channel conditions have significant changes, where the BS needs to rerun the reinforcement learning algorithm to obtain the new power allocation policies for each UE.

Proof of Property 1: Based on the UE's QoE utility definition and reward definition shown in (3) and (5), we obtain the reward expression of the ith UE:

$$r = \log_2\left(1 + B\log_2\left(1 + \frac{P_i \times |h_i|^2}{\sum_{k=i+1}^{N} P_k \times \left|h_k\right|^2 + \sigma^2}\right)\right) - c \times P_i. \quad (6)$$

It can be noticed from (6): for the Nth UE, as the strongest channel agent, its reward is only correlated with its own channel gain and the power allocated by the BS; for the UE_{N-1}, its reward is only affected by UE_N (the only UE who has stronger channel); for the UE_{N-2}, its reward is affected by UE_N and UE_{N-1} (the two UEs who have stronger channel than it); so on and so forth, the reward of UE_1 will be affected by all other UEs in the system. Thus, we conclude the optimal power allocation policies of UEs in the NOMA system follow the patterns:

(i) The optimal power allocation policy of UE_N is independent of other agents.
(ii) The optimal power allocation policy of UE_{N-1} will be only affected by the power allocation and channel of UE_N.
(iii) The optimal power allocation policy of UE_i will be affected by the power allocation and channel of UE_{i+1}, UE_{i+2},…, UE_N.
(iv) The optimal power allocation policy of UE_1 will be affected by all other UEs' power policy and channel conditions in the NOMA downlink system.

Based on the power allocation policies' patterns and Assumption 1 (where the channels remain constant for a short period of time), we conclude that the optimal power allocation policy for the NOMA downlink system would be reached one-by-one from UE_N to UE_1. That is to say, the MARL problem in our proposed system can be solved by seeking the optimal result for the *sequential* SARL problem.

Thus, instead of finding the equilibrium solution for the MARL system, where the tension among agents is always modelled either cooperative, competitive, or mixed game situations, we will study the conventional but sequential SARL in this paper, to find the optimal power allocation policy for each agent (in the decreasing sequence). Furthermore, our proposed sequential SARL method is treated as being centralised, where the BS runs the learning algorithms and obtains the optimal power allocation policies for all UEs in the system.

4.3 Q-learning approach

Recall that we use the model-free Q-learning method in our work, where both reward function and transition probability are unknown. To obtain the optimal action policy (i.e. power allocation strategy), which is denoted as

$$a_t = \arg\ \max\ Q(s_t, a_t), a_t \in \pi^*, \quad (7)$$

we have the following Bellman Q-function [18]

$$Q_{t+1}(s,a) = Q_t(s,a) + \alpha \times \left(r + \beta \max_{a' \in A} Q_t(s', a') - Q_t(s,a)\right) \quad (8)$$

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where α denotes the learning rate and β is the discount factor. It is worth pointing out that the value selection of β depends on the purpose of learning. In general, we assume $0 \leq \beta < 1$. We only take the immediate rewards into account if we set $\beta = 0$ or we consider the future rewards is more important than the immediate rewards if the value of β close to 1. It has been proved that in the MDP case, the Q-values would converge to the optimal Q^* if each action is executed in each state an infinite number of times on an infinite run with appropriately setting up of learning rate α [18, 19]. As illustrated in (7), we use the Q-learning method to update the Q value recursively. The Q value will be updated in each iteration based on the current Q value and the Q value from the next state by taking greedy action a'. To reach the optimal policy π^*, our proposed Q-learning method does not need to follow any specific policy (i.e. independent of the policy used during exploration), as long as we visit different states enough times.

4.4 Complexity analysis of the learning algorithms

In this paper, the process of obtaining an optimal power allocation policy includes two procedures. The first procedure describes the BS's centralised power allocation function, as shown in Algorithm 1 (see Fig. 3). The BS sets up the power levels that agents could explore in the learning procedure. Also, the BS defines the terminal states for agents' learning process based on the power budget in the first procedure. More specifically, the agent should stop exploring new solutions when the allocated power reaches edges (terminal states), i.e. $P_i \leq 0$ or $P_i \geq P_{\max}$. The second procedure shows that the BS runs the Q-learning for each UE to obtain the optimal power allocation policy, respectively.

The details of Q-learning algorithm for each UE (a.k.a. agent) is explained in Algorithm 2 (see Fig. 4), where the Epsilon greedy strategy is used. It is worth pointing out that the maximum iteration time K (lines 4 and 5) in the algorithm is determined by the convergence speed of the learning. The rationale behind this is that the BS would waste computing resource in the learning process if the $\boldsymbol{Q}$ matrix already converged. Thus, the value of K would be adopted around the iteration time that ensures $\boldsymbol{Q}$ matrix's convergence. The time complexity of running learning algorithm for all UEs is calculated as $O(NK)$. Owing to the size of the NOMA downlink system is relatively small, i.e. $N < 8$, plus the learning process of each agent would be brief with properly choosing of learning rate and exploration probability. The BS would complete the reinforcement learning very fast and obtain the global optimal power allocation policy efficiently. In our real simulation, we find the iteration that ensures the convergence of $\boldsymbol{Q}$ matrix is around 10,000–100,000 with properly set of α and ε (more details are shown in simulation section), even all starting points are randomly chosen.

5 Simulation

In this section, we carry out numerical simulations to evaluate the QoE utility performance of UEs under different situations, i.e. different channel conditions, power budgets, and the proposed sequential SARL method versus equal power allocation method in the NOMA downlink system. The channel/bandwidth information and other system set up parameters are shown in Table 1.

It can be noticed that factors such as channel gain and transmission power of UE would influence other UEs' data rates and QoE gain based on (2) and (3). In Fig. 5 we evaluate the QoE gain of UE under different UEs' channel conditions in the 2-UE system. We assume the BS' power budget is 30 dBm and is equally allocated to two UE. We observed that when the channel gain of UE is relatively high, i.e. the UE2 with channel gain 30 and 40 dB shown in the two top lines, it's QoE utility gain has slight improvement when the channel becomes better. While for the UE with the worse channel, i.e. the bottom two lines, the QoE gain would dramatically increase with the increase with its channel gain. In addition, the QoE gain of UE1 is also influenced by the channel of UE2 and the effects would be even severe when UE2's channel increases.

Initialization: Define the I/O of algorithm: Inputs: (1) The power budget $P_{\max}$ of the BS. (2) The channel gains of UEs. Outputs: (1) The global optimal power allocation policy of the BS.

BS managing procedure:

Initialize the state space and action space; Set the initial power allocation policy $(P_1, P_2, ..., P_N) = \overrightarrow{0}$

For k=N:1

If $\sum_{i=k+1}^{N} P_i < P_{\max}$

Perform the Q-learning procedure for UE_k and obtain the optimal power allocation policy for UE_k.

Else

The current power budget of BS is inadequate for all UEs in the system.

Return the current power policy for allocated UEs.

End for;

Output the global optimal power allocation policy.

Fig. 3 *Algorithm 1: the centralised power allocation algorithm of BS*

Initialization: Define the I/O of algorithm: Inputs: (1) The power allocation policies and channel gains of stronger UEs, i.e., $\forall UE_j, j > i$. (2) The UE_i's channel gain h_i. (3) The parameters for Q-learning process, e.g., learning rate α, discount factor β, and exploration probability ε. Outputs: (1) The optimal power allocation policy of UE_i.

Q learning procedure::

Initialize the Q matrix: Q = zeros(length(state), length(action)); The maximum number of iterations K. The initial state to start (randomly).

For s=1:K

X = rand(); Get one uniform random number;

X = sum(X>=cumsum([0, 1-ε, ε]).

If X==1

Next action ->choose exploitation;

Else

Next action ->choose exploration;

Based on reward definition $r_t = U_i|_{h_{t+1}, I_{t+1}, R_{t+1}, P_{t+1}}$ and Equation (6), calculate the reward of agent at next action. Update the state;

Update the Q matrix using the Q-learning rule: $Q_{t+1}(s,a) = Q_t(s,a) + \alpha \times \left(r + \beta \max_{a' \in A} Q_t(s', a') - Q_t(s,a) \right)$

If current state is terminal state

Restart the episode with a new (random) state;

End for

Output the optimal action policy based on the Q matrix;

Fig. 4 *Algorithm 2: the learning procedure of agent UE_i*

Table 1 Parameters used in the simulations

Symbol	Range of value
h_i	0–40 dB
N	2–5
σ^2	0 dB
B	800 MHz
$P_{\max}$	23–38.5 dBm
c	1
α, β, ε	0.1–1

In Figs. 6 and 7, we test the data rate and QoE performance of UEs under fixed power allocation policies in the 2-UE system. The channel gains of two UEs are set as 20 and 30 dB. The power ratio in this simulation denotes the power of UE1 to the power of UE2, i.e. 1:2 means the power of UE1 is half of UE2's, 1:1 means two UEs equally share the power, and 2:1 means the UE1's power is two times of UE2's. We can observe from Fig. 6 that the data rates of UEs have big differences under three power allocation policies. Specifically, the UE1 would have the best data rate performance under the 2:1 scheme, while the UE2 is on the contrary (who has the best data rate under 1:2 scheme). We conclude that when the

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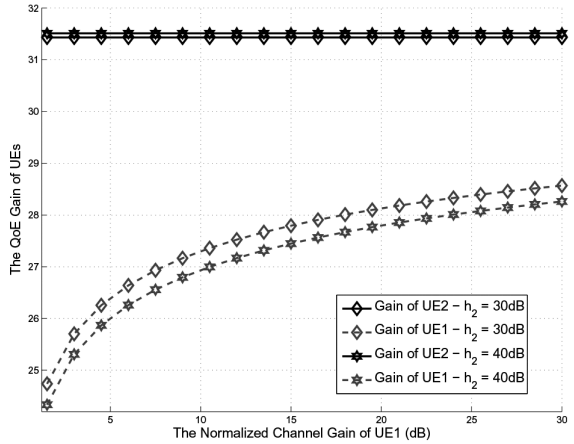


Fig. 5 *In the 2-UE NOMA downlink system, the QoE performance of UEs under different channel conditions*

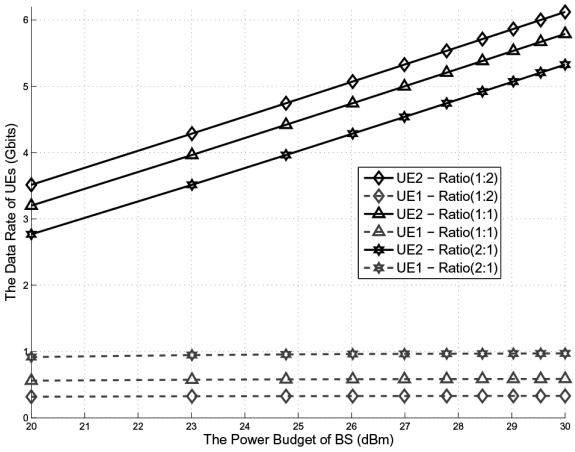


Fig. 6 *Illustration of the data rate of UEs in the 2-UE NOMA downlink system with increasing BS' power budget*

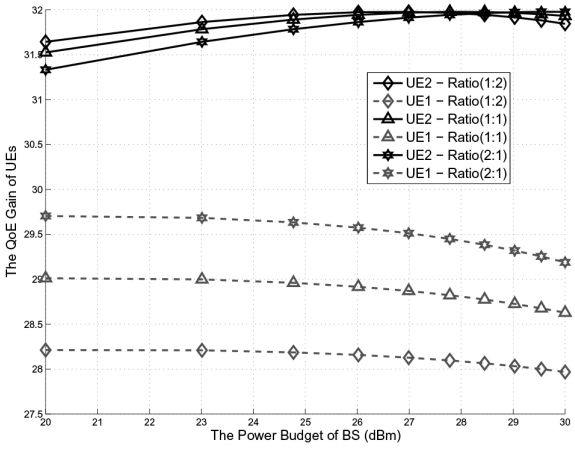


Fig. 7 *QoE performance evaluation under different power allocation policies*

power budget is fixed, we can control the data rates of UEs in the NOMA system by properly allocating different powers for UEs. In addition, with the increasing of BS' power budget, UE2's data rate gets significantly improvement while UE1's data rate only obtains slight improvement. The reason behind such a phenomenon is that the UE1 will experience interference from UE2 in the NOMA downlink. The interference becomes even severe with the increase of UE2's power. It is worth pointing out that such a scheme is suitable for some internet of things scenarios where certain devices only need to be served with low data rates.

Fig. 7 illustrates the QoE gain of two UEs in these three power allocation policies. Similar to the data rate performance, we see that the 2:1 power allocation would benefit UE1 the most when the power budget of BS is fixed. While the QoE gains of UE1 have bigger variation comparing with the UE2's in different power allocation policies. We also observed that the QoE of UE2 will improve with the increase of power at first and start to slowly decrease after reaching the maximum point. That is to say, the QoE of UE2 has concave property. The cause of the concavity of UE2's QoE is that the cost portion (shown in (3)) will increase when the power keeps growing up, and eventually the cost part will cause the QoE to go down. We find that the QoE of UE1 is concave too if we keep the power of UE2 as constant. It is important to point out that our proposed SARL method will lead the BS reaching the optimal power allocation policies for all UEs, where the optimal powers exactly ensure all UEs obtain the maximum QoE.

Next, we evaluate how the model parameters (i.e. learning rate and exploration probability) would affect the iteration time (the minimum of steps of convergence) the learning process. In our simulation, we noticed that comparing with the learning rate α, the exploration probability ε has bigger influence on the convergence speed. More specifically, there are 93 states in our simulation (including two terminal states), it would take more than 1×10^{10} times to see the convergence of power allocation policy when $\varepsilon < 0.7$ and take 10^8–10^9 time when $\varepsilon = 0.7$, even we set $\alpha = 1$. Thus, we present our results with $\varepsilon = 0.8 - 0.95$ and various values of α, as shown in Fig. 8. We observed that the convergence speed will dramatically increase (i.e. needs less iterations) with the increasing of exploration probability. The rationale behind this is that: there are two random steps in the epsilon greedy Q-learning algorithm, it would take lots of (i.e. billions) tries to explore all states with small ε. On the contrary, the agent would explore all states and reach the convergence power allocation policy very fast when ε is relatively high, i.e. <10,000 times with $\varepsilon = 0.95$. We also observed from Fig. 8 that with the increasing of the learning rate, the algorithm needs less iteration times to reach the convergence.

Fig. 9 shows the UE's QoE performance of two different power allocation policies. We consider two UEs in the system and $|h_1|^2 = 5$ dB, $|h_1|^2 = 10$ dB. The parameters for learning are set as $\alpha = 0.5$, $\beta = 0.5$, and $\varepsilon = 0.9$. The equal scheme shown in the solid lines implies two UEs equally share the BS's transmission power. The SARL scheme shown in dash lines implies the system will take the power allocation policy from the output of the reinforcement learning. It can be noted from the result that the QoE gain of UE2 is better than UE1 in both policies. This is due to UE1 has smaller channel gain in the 2-UE system, it would experience the interference from strong UE (i.e. UE2). In addition, we can see that the QoE gain of UE1 in the SARL would dramatically increase from 0 to 26 when the BS's power budget is >26 dBm. The rationale behind this is: in our proposed power allocation method, the BS would only serve the strong UE if the power budget is insufficient (as shown in line 8–9 in Algorithm 1). So the UE1 keeps 'starving' before the BS's power budget reaches 26 dBm. The equal power allocation policy works better when the BS is limited by the power supply. While with the increasing of the BS's power, the proposed SARL policy will surpass the equal scheme. Besides, due to the mutual interference will increase with the increasing power of strong UE, the QoE gain of UE1 starts going down after the BS' power budget >28 dBm in the equal scheme.

Next, we evaluate the QoE performance of each UE in the 3-UE NOMA downlink system. We set the channel gains of three UEs as 5, 10, and 20 dB. The result is shown in Fig. 10. As we can see that UE3 has the best QoE performance due to its highest channel gain. Along with the increasing of BS' power, the QoE gains of UE2 and UE1 will increase first and gradually decrease later. The reason for the decrease is that UE2 and UE1 will experience the raising interference from strong UE(s). It is worth pointing out that our proposed SARL scheme will benefit the weak UEs when the BS has enough power.

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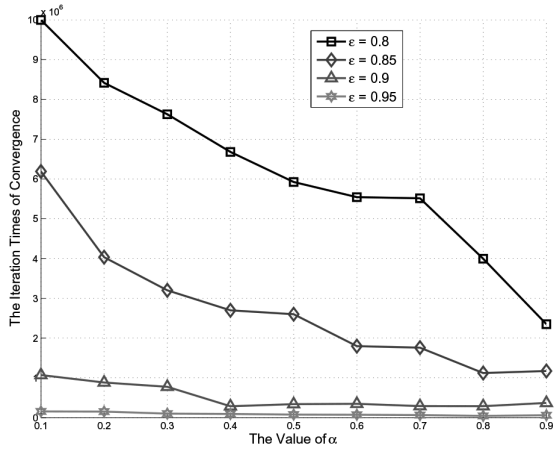


Fig. 8 *Illustration of convergence speed of algorithm under the different system set up parameters. We consider the learning rate α and the exploration probability ε in this simulation*

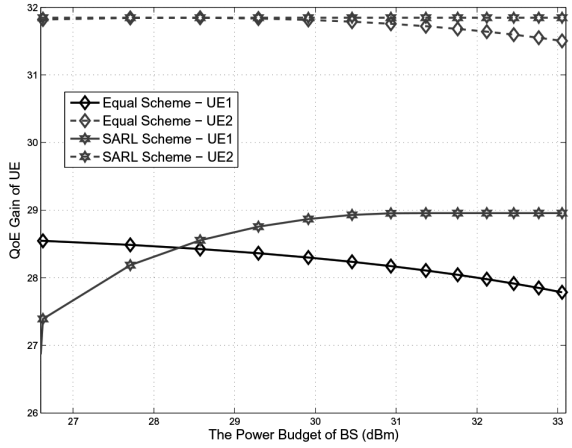


Fig. 9 *QoE performance of UEs in the 2-UE NOMA downlink system, we consider two power allocation schemes in this simulation, the equal power allocation scheme and the allocation scheme from SARL*

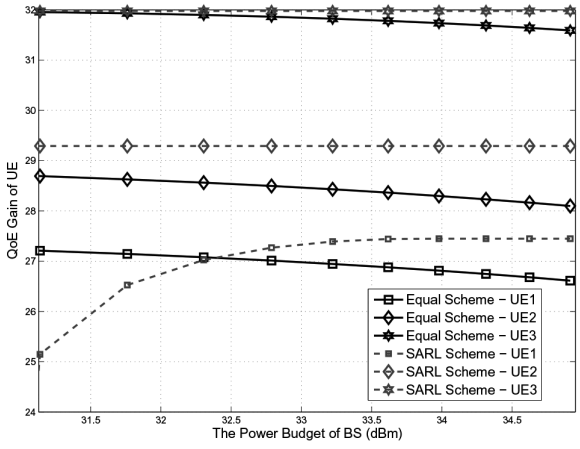


Fig. 10 *QoE gain of UEs in the 3-UE NOMA downlink system. The channel gains of three UEs follow* $|h_1|^2 \le |h_2|^2 \le |h_3|^2$

Furthermore, we evaluate the QoE performance in 4-UE and 5-UE NOMA downlink system. The channel gains of UE are set as [5, 10, 20, 30] for the 4-UE system and [5, 10, 20, 30, 40] for the 5-UE system respectively. The average QoE gain of UE versus power budget simulation results is shown in Fig. 11. It can be noted in multiple UEs NOMA downlink system (i.e. $N > 3$), the average QoE of each UE (a.k.a the over system QoE performance) keeps

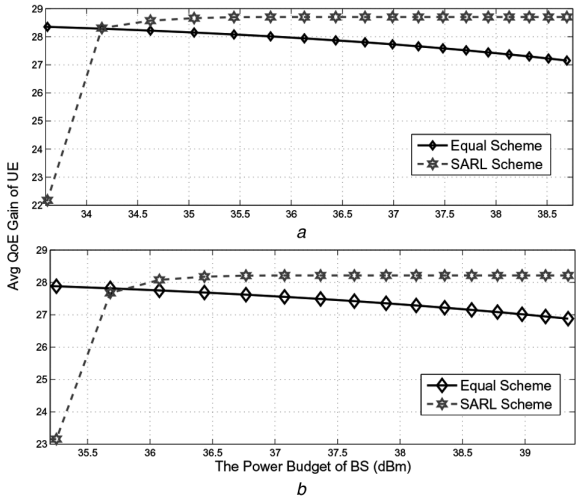


Fig. 11 *Average QoE gain of UE in*
(a) 4-UE NOMA system, ***(b)*** 5-UE NOMA system

decreasing with the increasing of power budget in the equal power allocation scheme. There are two reasons for such a result: (i) with the increasing of transmission power for all UEs, the weak UEs would experience more interference from strong UEs due to the bigger power, even the strong UEs can perform SIC to cancel the interference from weak UEs. This would eventually cause a decreasing in QoE gain. (ii) As the QoE definition is shown in (3), the power cost of each UE is linearly increasing with the changing power. The non-stop increase of cost would eventually cause a decrease in QoE gain. We also observed that with the increase of power budget, the average QoE gain of UE in the SARL scheme keeps growing and outperformance the equal allocation scheme. In general, the power budget of BS is sufficient to support all the UEs it served. So we conclude the proposed SARL has a better performance compared with the conventional preset fixed power allocation policy (i.e. equal power allocation).

6 Conclusion

Aiming at improving the UEs' QoE utilities in the wireless multimedia service, a reinforcement Q-learning power allocation approach is developed to approximate the optimal power allocation policy of BS in the NOMA downlink scenario. First, to quantitatively analyse user's perceived QoE, we adopted a light-weighted two-level logarithmic mathematical QoE model, where the channel conditions, mutual interference, and UE's achievable data rate are jointly considered. Then, we proposed a sequential SARL method to address the MARL problem. Through such a method the BS would approximate the optimal power allocation policy for each UE during the sequential Q-learning processes, in a unique descending order in terms of UE channel gains. Finally, numerical simulation results demonstrated the enhanced QoE utility performance of the proposed SARL scheme in the NOMA downlink system, especially when the BS has enough power budget.

A central challenge of the reinforcement learning application in the NOMA wireless multimedia is the definition of an appropriate formal goal for the multi-agent learning system. The relationships among UEs could either be non-cooperative (i.e. maximising their individual QoE goal) or be cooperative (i.e. improving the overall system power allocation efficiency). Our future work would focus on how to better define the goal by leveraging techniques such as game theory and optimal control theory.

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